



Devoir

Exercice 1.

Montrer, à partir de la définition de limite, que $\lim_{n \rightarrow +\infty} \frac{6n+1}{2n+3} = 3$

Exercice 2.

Soit (u_n) la suite réelle définie par :
$$\begin{cases} u_1 = 0 \\ u_{n+1} = \frac{7u_n + 4}{3u_n + 3} \end{cases}$$

- 1) Calculer u_2 et u_3
- 2) Montrer par récurrence que : $0 \leq u_n \leq 2$ pour tout $n \geq 1$.
- 3) Montrer que (u_n) est croissante
- 4) En déduire que (u_n) converge et déterminer sa limite.

Exercice 3.

Calculer les limites des suites suivantes :

$$u_n = \frac{1}{n^2 + k^2} ; \quad v_n = \frac{n^3}{3} ; \quad w_n = \sqrt{4n^2 + 8n + 9} - 2n - 3$$

Exercice 4.

Calculer les limites suivantes :

$$1) \lim_{x \rightarrow 4} \frac{\sqrt{1+2x} - 3}{\sqrt{x} - 2} ;$$

$$2) \lim_{x \rightarrow 0} \left(\frac{e^{2x} - 1}{e^{3x}} \right) ;$$

$$3) \lim_{x \rightarrow 0} \frac{\sin 4x}{\tan 5x}$$

Exercice 5.

Etudier la continuité des fonctions suivantes sur leurs domaines :

$$1) f(x) = \begin{cases} \frac{\sin x}{\sqrt{1+x^2} - 1 - x} & \text{si } x \neq 0 \\ 0 & \text{si } x = 0 \end{cases} ;$$

$$2) f(x) = \begin{cases} \frac{x - |x|}{x} & \text{si } x \neq 0 \\ 2 & \text{si } x = 0 \end{cases}$$

Exercice 6. Peut-on prolonger la fonction f par continuité au point 1? Si oui, donner l'expression du prolongement :

$$f(x) = \frac{\sqrt{x^2 + x + 2} - 2}{x - 1}$$

Exercice 01



$$\frac{n+1}{n+3} = 3$$

$\forall \varepsilon > 0 \exists n_\varepsilon > \varepsilon$ tel que

$$|u_n - l| < \varepsilon$$

$$\left| \frac{6n+1}{2n+3} - 3 \right| < \varepsilon$$

$$\Rightarrow \left| \frac{6n+1-6n-9}{2n+3} \right| < \varepsilon$$

$$\Rightarrow \left| \frac{-8}{2n+3} \right| < \varepsilon$$

$$\Rightarrow \frac{8}{2n+3} < \varepsilon$$

$$\frac{2n+3}{8} > \frac{1}{\varepsilon}$$

$$\varepsilon(2n+3) > 8$$

$$\Rightarrow 2n+3 > \frac{8}{\varepsilon}$$

$$2n > \frac{8}{\varepsilon} - 3$$

$$\Rightarrow n > \left(\frac{8}{\varepsilon} - 3 \right) \times \frac{1}{2}$$

donc il exist $n_3 = \left(\frac{8}{\varepsilon} - 3 \right) \times \frac{1}{2}$

donc

$$\lim_{n \rightarrow \infty} \frac{6n+1}{2n+3} = 3$$

Exercice 03

$$b_n = \frac{n^3}{3^n}; \text{ On a } a = e$$

$$= \frac{3 \ln(n)}{e}$$

$$n \ln(3)$$

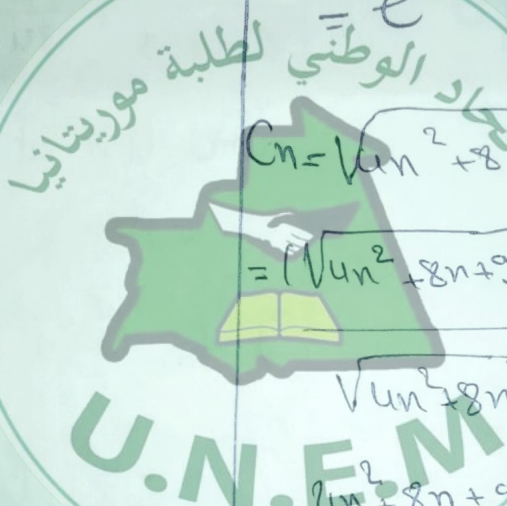
$$= 3 \ln(n) - n \ln(3)$$

$$e_n \left(3 \frac{\ln(n)}{n} - \ln(3) \right)$$

$$= e$$

$$- n \ln(3) \rightarrow -\infty$$

$$= e = 0$$



$$C_n = \sqrt{4n^2 + 8n + 9} - 2n - 3$$

$$= \frac{(\sqrt{4n^2 + 8n + 9} - (2n+3))(\sqrt{4n^2 + 8n + 9} + (2n+3))}{\sqrt{4n^2 + 8n + 9} + (2n+3)}$$

$$\sqrt{4n^2 + 8n + 9} + (2n+3)$$

$$= \frac{4n^2 + 8n + 9 - (2n+3)^2}{\sqrt{4n^2 + 8n + 9} + (2n+3)}$$

$$\sqrt{4n^2 + 8n + 9} + (2n+3)$$

$$= \frac{4n^2 + 8n + 9 - (4n^2 + 12n + 9)}{\sqrt{4n^2 + 8n + 9} + (2n+3)}$$

$$\sqrt{4n^2 + 8n + 9} + (2n+3)$$

$$= \frac{4n^2 + 8n + 9 - 4n^2 - 12n - 9}{\sqrt{4n^2 + 8n + 9} + (2n+3)}$$

$$\sqrt{4n^2 + 8n + 9} + (2n+3)$$

$$= -4n$$

$$\times \left(\sqrt{4 + \frac{8}{n} + \frac{9}{n^2}} + \left(2 + \frac{3}{n} \right) \right)$$

$$= -2$$



$$= \frac{-4}{\sqrt{4} + 2} = \frac{-4}{2} = -2$$

$$\lim_{n \rightarrow \infty} \sqrt{4n^2 + 8n + 4} - 2n - 3 = -1$$

Exercice 2

$$f(x) = \frac{x^2}{1+x^2}$$

$$u_0 = 0$$

$$u_{n+1} = \frac{7u_n + 4}{3u_n + 3}$$

$$u_1 = \frac{7(u_0) + 4}{3(u_0) + 3}$$

$$u_1 = \frac{7(0) + 4}{3(0) + 3} = \frac{4}{3}$$

$$u_2 = \frac{4}{3}$$

$$u_3 = \frac{7 \times \frac{4}{3} + 4}{3 \times \frac{4}{3} + 3} = \frac{\frac{28}{3} + 4}{4 + 3} = \frac{\frac{40}{3}}{7} = \frac{40}{21}$$

$$u_3 = \frac{40}{21}$$

2) Pour $n = 1$

$$0 \leq u_1 \leq 2$$

Pour $n = k$ On a

$$0 \leq u_k \leq 2$$

Pour $n+1$ mg $0 \leq u_{k+1} \leq 2$

$$0 \leq u_n \leq 2$$

$$0 \leq 7u_n \leq 14$$

$$\frac{4}{3u_n + 3} \leq \frac{7u_n + 4}{3u_n + 3} \leq \frac{18}{3u_n + 3}$$

donc u_n est compris entre

$$0 \leq u_n \leq 2$$

3) $u_{n+1} - u_n$

$$\frac{7u_{n+1} + 4}{3u_{n+1} + 3} - u_n =$$

$$\frac{7u_n + 4 - 3u_n^2 - 3u_n}{3u_n + 3}$$

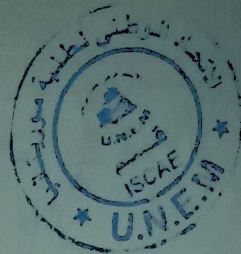
$$= \frac{-3u_n^2 + 4u_n + 4}{3u_n + 3}$$

$$3u_n + 3 \rightarrow (+)$$

$$-3u_n^2 + 4u_n + 4 = 0$$

$$16 + 48 = 64$$

Devoir 2021-2022



$$b^n = \frac{n^3}{n^3}; \text{ On a } a^b = e^{b \ln a}$$

$$= \frac{3 \ln(n)}{n \ln(3)}$$

$$n(3^{\frac{\ln(n)}{n}} - \ln(3))$$

$$= 3 \ln(n) - n \ln(3) = e$$

$$e^{-n \ln(3)} \boxed{e^{-\infty} = 0}$$

$$= e$$

$$C_n = \sqrt{4n^2 + 8n + 9} - 2n - 3$$

$$= \frac{(\sqrt{4n^2 + 8n + 9} - (2n + 3))(\sqrt{4n^2 + 8n + 9} + (2n + 3))}{(\sqrt{4n^2 + 8n + 9} + (2n + 3))}$$

$$= \frac{4n^2 + 8n + 9 - (4n^2 + 12n + 9)}{\sqrt{4n^2 + 8n + 9} + (2n + 3)}$$

$$= \frac{4n^2 + 8n + 9 - 4n^2 - 12n - 9}{\sqrt{4n^2 + 8n + 9} + (2n + 3)}$$

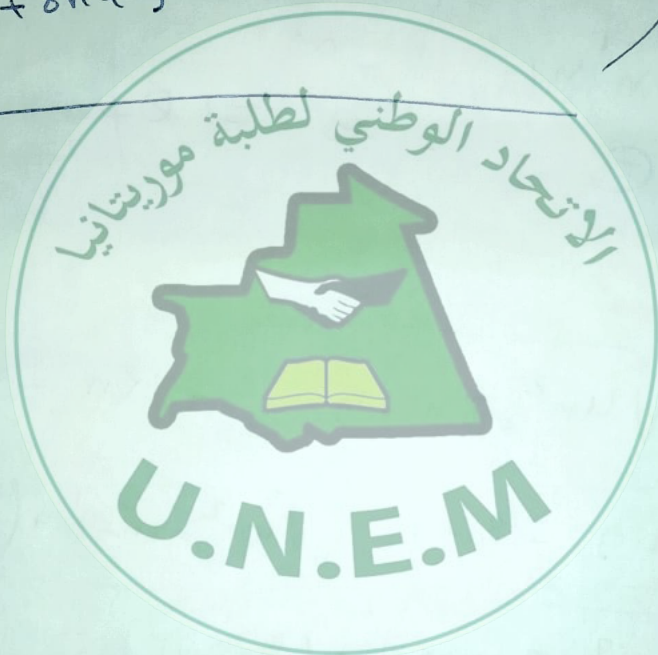
$$= \frac{-4n}{n(\sqrt{4 + \frac{8}{n} + \frac{9}{n^2}}) + (2 + \frac{3}{n})}$$



$$\frac{1}{n} + \frac{9}{n} + \left(2 + \frac{3}{n}\right)$$

$$= \frac{-4}{\sqrt{n} + 2} = \frac{-4}{u} = -1$$

$$u = \sqrt{4n^2 + 8n + 9} - 2n - 3 \dots - 1$$



Devoir 2021-2022



$$\lim_{n \rightarrow 4} \frac{(1+2n-3)}{\sqrt{n}-2}$$

$$1) \lim_{n \rightarrow 4} \frac{(\sqrt{1+2n}-3)(\sqrt{1+2n}+3)(\sqrt{n}+2)}{(\sqrt{n}-2)(\sqrt{1+2n}+3)(\sqrt{n}+2)}$$

$$= \lim_{n \rightarrow 4} \frac{(1+2n-9)(\sqrt{n}+2)}{(\sqrt{1+2n}+3)(n-4)}$$

$$= \frac{2n-8(\sqrt{n}+2)}{(\sqrt{1+2n}+3)(n-4)}$$

$$\lim_{n \rightarrow 4} f(n) = \frac{2(n-4)(\sqrt{n}+2)}{(n-4)(\sqrt{1+2n}+3)}$$

$$= \frac{2(n-4)(\sqrt{n}+2)}{(\sqrt{1+2n}+3)(n-4)}$$

$$\lim_{n \rightarrow 4} g(n) = \frac{2(n-4)(\sqrt{n}+2)}{(n-4)(\sqrt{1+2n}+3)}$$

$$= \frac{2(\sqrt{n}+2)}{\sqrt{1+2n}+3}$$

$$\Rightarrow \lim_{n \rightarrow 4} f(n) = \frac{2\sqrt{4}+2}{\sqrt{1+2 \times 4}+3}$$





$$\frac{2(4)}{\sqrt{3} + 3} = -\frac{8}{6}$$

$$\frac{e^a}{e^b} = e^{a-b}$$

$$2 - \lim_{n \rightarrow \infty} \frac{e^{\frac{6n^2}{2n-1}}}{e^{3n}}$$

$$= \lim_{n \rightarrow \infty} \frac{6n^2}{e^{2n-1} - 3n}$$

$$= \lim_{n \rightarrow \infty} e^{\frac{6n^2 - 3n(2n-1)}{2n-1}}$$

$$= \lim_{n \rightarrow \infty} e^{\frac{12n^2 - 6n}{2n-1}}$$

$$= \lim_{n \rightarrow \infty} e^{\frac{12n}{2 - \frac{1}{n}}}$$

$$= \lim_{n \rightarrow \infty} e^{\frac{12}{2 - \frac{1}{n}}}$$

$$= \lim_{n \rightarrow \infty} 4n = e^{\frac{12}{2}}$$

Devoir 2021 - 2022



$$1) f(x) = \begin{cases} \frac{\sin x}{\sqrt{1+x^2} - 1 - x} & \sin x \neq 0 \\ 0 & \sin x = 0 \end{cases}$$

$$= \frac{(\sin x)(\sqrt{1+x^2} + (1+x))}{(\sqrt{1+x^2} - (1+x))(\sqrt{1+x^2} + (1+x))}$$

$$= \frac{\sin x (\sqrt{1+x^2} + (1+x))}{1 + x^2 - (1+x)^2}$$

$$= \frac{\sin x (\sqrt{1+x^2} + (1+x))}{1 + x^2 - 1 - 2x - x^2}$$

$$= \frac{\sin x (\sqrt{1+x^2} + (1+x))}{-2x}$$

$$= \frac{\cancel{x} \left(\frac{\sin x}{x} \sqrt{1+x^2} + (1+x) \right)}{-2\cancel{x}}$$

$$= \frac{1(\sqrt{1+0} + 1)}{-2} = \frac{1+1}{-2} = \frac{2}{-2} = -1$$

$$\lim f(x) \neq f(0)$$

Donc f n'est pas Continue





$$\lim_{n \rightarrow \infty} \left(\frac{n-|n|}{n} \right) = \begin{cases} \frac{n-|n|}{n} & n \neq 0 \\ 2 & n = 0 \end{cases}$$

$$\frac{n}{n} + \frac{n}{n} \lim_{n \rightarrow 0} < 0$$

$$2 \lim_{n \rightarrow 0} = 0$$

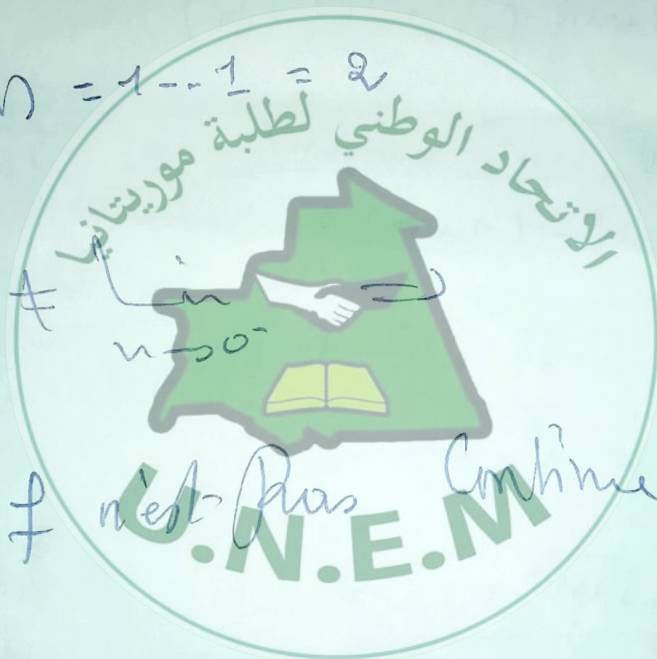
$$\frac{n}{n} - \frac{n}{n} \lim_{n \rightarrow 0} > 0$$

$$\lim_{n \rightarrow 0^+} f(n) = 1 - 1 = 0$$

$$\lim_{n \rightarrow 0^-} f(n) = 1 - \frac{1}{1} = 0$$

$$\lim_{n \rightarrow 0^+} \neq \lim_{n \rightarrow 0^-}$$

donc f n'est pas continue



Exercice 06 Devoir



$$\frac{\frac{n^2+n+2-2}{n-1}}{(\sqrt{n^2+n+2}-2)(\sqrt{n^2+n+2}+2)}$$

$$(n-1)(\sqrt{n^2+n+2}+2)$$

$$= \frac{n^2+n+2-4}{(n-1)(\sqrt{n^2+n+2}+2)}$$

$$= \frac{n^2+n-2}{(n-1)(\sqrt{n^2+n+2}+2)}$$

$$= \frac{(n-1)(n+2)}{(n-1)(\sqrt{n^2+n+2}+2)}$$

$$= \frac{n+2}{\sqrt{n^2+n+2}+2}$$

$$\lim_{n \rightarrow 1} f(n) = \lim_{n \rightarrow 1} \frac{n+2}{\sqrt{n^2+n+2}+2}$$

$$= \frac{1+2}{\sqrt{1+1+2}+2}$$

$$= \frac{3}{\sqrt{4}+2} = \frac{3}{4}$$

1 et 2

$$\lim_{n \rightarrow 1} f(n) = \frac{3}{4}$$

donc f est prolongable par continuité

$$\tilde{f} = \begin{cases} \frac{\sqrt{n^2+n+2}-2}{n-1} & \text{si } n \neq 1 \\ \frac{3}{4} & \text{si } n = 1 \end{cases}$$

